Statistics & Probability

ANSWER KEY

Section 1: Counting Principles & Probability

1 A password consists of 3 letters followed by 2 digits. If repetition of letters is NOT allowed, how many such passwords can be created?

a. $1,560,000$ ✓

b. $1,757,600$

c. $1,404,000$

d. $2,600,000$

Letter arrangements: $26 \times 25 \times 24 = 15,600$. Digit arrangements: $10 \times 10 = 100$. Total $= 15,600 \times 100 = 1,560,000$.

2 Five people are to be seated in a row of five chairs. What is the probability that two specific people, Alice and Bob, sit next to each other?

a. $\frac{2}{5}$ ✓

b. $\frac{1}{5}$

c. $\frac{1}{2}$

d. $\frac{3}{5}$

Treat Alice and Bob as a single block: $4! \times 2! = 24 \times 2 = 48$. Total arrangements $= 5! = 120$. Probability $= \frac{48}{120} = \frac{2}{5}$.

3 What is the probability of obtaining exactly 3 heads in 5 tosses of a fair coin?

a. $\frac{5}{16}$ ✓

b. $\frac{1}{2}$

c. $\frac{3}{16}$

d. $\frac{5}{32}$

Binomial probability: $\binom{5}{3}(\frac{1}{2})^3(\frac{1}{2})^2 = 10 \times \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$.

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Section 2: Conditional Probability & Independence

4 If $P(A) = 0.4$, $P(B \mid A) = 0.75$, and $P(B) = 0.5$, what is $P(A \mid B)$?

a. 0.60 ✓

b. 0.50

c. 0.30

d. 0.75

$P(A \cap B) = P(A)P(B \mid A) = 0.4 \times 0.75 = 0.3$. Then $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.60$.

5 Two events A and B satisfy $P(A) = 0.3$ and $P(B) = 0.6$. If A and B are independent, what is $P(A \cup B)$?

a. 0.72 ✓

b. 0.90

c. 0.78

d. 0.18

$P(A \cap B) = 0.3 \times 0.6 = 0.18$. $P(A \cup B) = 0.3 + 0.6 - 0.18 = 0.72$.

6 In a certain school, 60% of students take French, 40% take Spanish, and 20% take both. What is the probability that a student takes French given that they take Spanish?

a. 0.50 ✓

b. 0.33

c. 0.60

d. 0.25

$P(\text{French} \mid \text{Spanish}) = \frac{P(\text{French} \cap \text{Spanish})}{P(\text{Spanish})} = \frac{0.20}{0.40} = 0.50$.

Section 3: Normal Distributions & Standard Scores

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7 In a standard normal distribution $Z \sim N(0, 1)$, what is $P(-1.96 \leq Z \leq 1.96)$?

a. 0.95 ✓

b. 0.68

c. 0.997

d. 0.90

By standard normal probability tables, 95% of the distribution falls within 1.96 standard deviations of the mean.

8 A dataset has mean $\mu = 100$ and variance $\sigma^2 = 25$. What score corresponds to a z-score of 1.8?

a. 109 ✓

b. 105

c. 118

d. 145

Standard deviation $\sigma = \sqrt{25} = 5$. Score $x = \mu + z\sigma = 100 + 1.8(5) = 100 + 9 = 109$.

9 If 16% of scores on a normal test are above 84 and the mean is 76, what is the standard deviation?

a. 8 ✓

b. 4

c. 12

d. 16

In a normal distribution, the top 16% corresponds to approximately 1 standard deviation above the mean ($\mu + \sigma$). Thus $76 + \sigma = 84 \Rightarrow \sigma = 8$.

Section 4: Regression & Correlation

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10 If the correlation coefficient between two variables is $r = -0.85$, what does this indicate?

a. **A strong negative linear association** ✓

b. A weak negative linear association

c. A strong positive linear association

d. No linear association

Values of r near -1 indicate a strong negative linear relationship between the two variables.

11 In a least-squares linear regression line $\hat{y} = 2.5x + 14$, what is the predicted change in y for each unit increase in x?

a. $+2.5$ ✓

b. $+14$

c. $+5.6$

d. Cannot be determined

The slope 2.5 represents the estimated change in the response variable y for each 1-unit increase in the explanatory variable x.

12 An observed data point has coordinates $(4, 25)$. If the regression model predicts $\hat{y} = 23$ for $x = 4$, what is the residual?

a. $+2$ ✓

b. -2

c. $+48$

d. $+0.08$

Residual $= y - \hat{y} = 25 - 23 = +2$.

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